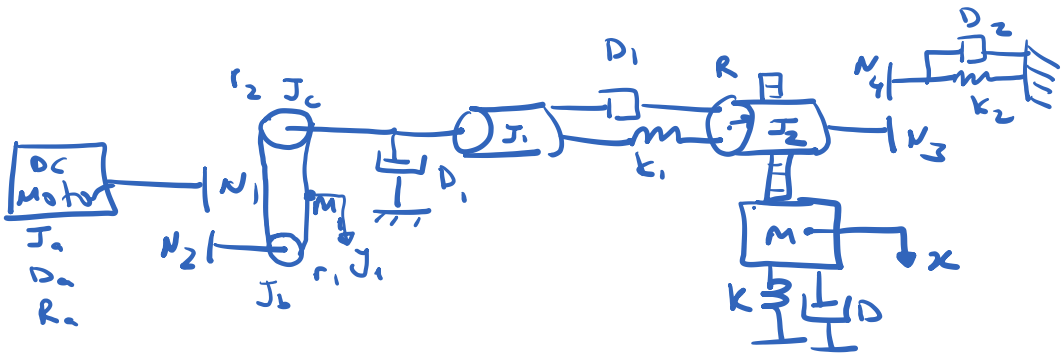
MKT5106 Feedback Control

Name:_____

ID:_____

Take-Home Final

	I	II.1	II.2	II.3	II.4	II.5



1) MODELLING: [20 pts.] Given the mechanical system shown in the schematic above:

   1. Derive the block diagram representation of the system.
   2. Develop the corresponding state-space model.
   3. Using MATLAB, obtain the transfer function of the system by setting all physical parameters equal to 1.

   The system consists of:
   Conveyor belt parameters: r_1, r_2 radii of the discs, J_1, J_2 inertias, m_1 the mass of the rigid belt. Rack-n-pinion parameters: R radius of the drum, J_2 its inertia, m, K, D mass and translational spring and damping. Motor parameters: k_b, k_t, R_a, J_a, D_a are considered. $e_a(t)$ DC voltage applied, $x(t)$ displacement of the translational port of the rack-n-pinion.

2) CONTROL: [85 Pts.] A dynamical system is modeled by

$$G(s) = \frac{10(s+3)}{(s^2+0.4s+8.96)(s+1)}$$

   a) [5 pts.] Analyze its transient and steady state characteristics to a unit step input.

   Our objective is to design a negative feedback controller to

   - Reduces the settling time by a factor of three
   - Limits the percent overshoot to at most 5%
   - Achieves zero steady-state error for reference tracking of step signal.

   b) [20 pts.] Using root-locus, design a controller, denoted as C1.
   c) [20 pts.] For the same system design a state feedback controller, say C2. Provide a block diagram of the resulting closed-loop control system .
   d) 20 pts.] Assume that only the phase variable state vector's first element is measurable as the output, y. Determine whether the system is observable and controllable first. Then, design an observer-based output feedback controller, say C3.
   e) [5 pts.] Compare the three controllers (C1, C2, and C3). Which controller would you choose for implementation, and why?
   f) [15 pts.] Derive the difference equations for the selected (best) controller when implemented on a digital processor with a control loop sampling period of 50 ms.
   Additionally, explain what precautions should be taken at the system input to prevent sampling-induced artifacts, such as aliasing or noise amplification.

   MUST verify all answers in Matlab or Python